

Improving Enterprise Data Product Discoverability through AI-Driven Metadata Enrichment: A Real-World Case Study

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Keywords

computing and processing, data product discoverability, enterprise data management, large language models, metadata enrichment, semantic retrieval, sentence-bert

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Enterprise data catalogs increasingly rely on metadata to support data product discoverability, yet metadata quality and semantic misalignment remain major obstacles in real-world settings. This study investigates how artificial intelligence techniques can be applied incrementally to improve metadata-driven data discovery under realistic enterprise constraints. We propose a step-by-step experimental framework that progressively introduces business glossary expansion, semantic retrieval using Sentence-BERT, LLM-generated data product descriptions, synthetic query generation, model fine-tuning, and LLM-based reranking, starting from a traditional keyword-based baseline. The framework is evaluated on a real-world enterprise metadata dataset using consistent, industry-aligned retrieval metrics. The results show that the transition from lexical to semantic retrieval yields the largest observed performance gains, substantially improving both precision and coverage. Metadata enrichment techniques provide incremental benefits, while more advanced approaches exhibit diminishing returns and strong dependence on data quality. The findings further indicate that task-aligned, fit-for-purpose models can outperform larger, general-purpose alternatives in enterprise metadata retrieval. Overall, the proposed framework offers practical guidance for incremental AI adoption in enterprise data discovery and is designed to be adaptable across organizations with different levels of data maturity, resources, and technical expertise.

Key words: Data product discoverability · Metadata enrichment · Semantic retrieval · Sentence-BERT · Large language models · Enterprise data management

1 Introduction

As organizations increasingly adopt data-centric operating models, data is no longer viewed as just one more part of business processes but as a reusable asset that supports analytics, automation, and decision-making [1]. This shift has led to the emergence of concepts such as Data Products [2], where datasets are treated as products with defined ownership, documentation, and quality expectations. Despite these advances, data discoverability remains a persistent challenge in many enterprise environments.

In practice, data users often struggle to identify relevant datasets even when these datasets already exist and are technically accessible through centralized platforms or data catalogs. A key contributing factor to this problem is the quality of metadata. Metadata is frequently incomplete, overly technical, inconsistent across domains, or outdated, limiting its usefulness for search and discovery [3]. As a result, organizations experience duplicated analytical effort, low data reuse, and reduced trust in data assets.

Traditional data discovery mechanisms rely primarily on keyword-based retrieval techniques, such as TF-IDF, which match user queries to metadata based on lexical similarity. While these methods are simple, interpretable, and computationally efficient, they are fundamentally limited in their ability to capture semantic meaning. This limitation becomes particularly evident in enterprise settings, where user queries vary significantly in phrasing and intent, and metadata often contains abbreviations or system-oriented terminology that does not align with user language.

The experimental results of this study confirm these limitations. When applied to a real-world enterprise dataset, the keyword-based baseline retrieved the correct data product as the top result in only about half of the user queries, and failed to surface the correct result within the top five recommendations in more than one third of the cases. These findings illustrate that purely lexical approaches are insufficient for supporting effective data discovery in complex organizational environments.

Recent advances in Artificial Intelligence offer promising alternatives. Semantic embedding models, such as Sentence-BERT (SBERT), enable context-aware similarity matching between user queries and metadata descriptions. In addition, Large Language Models (LLMs) can be leveraged to enrich metadata by generating user-friendly dataset descriptions, expanding technical terms through business glossaries, or producing synthetic user queries that approximate real search behavior.

While these techniques have been studied individually, there is limited empirical evidence on how they perform when applied incrementally within a single retrieval pipeline. Existing studies often focus on isolated components, such as semantic search or synthetic query

generation, without examining their cumulative impact or the trade-offs introduced by increasing system complexity. As a result, organizations lack practical guidance on how to transition from traditional retrieval methods to more advanced AI-driven approaches in a controlled and measurable way.

This paper addresses this gap by proposing a step-by-step experimental framework for improving data product discoverability through progressive metadata enrichment. Starting from a traditional TF-IDF baseline, the framework incrementally introduces business glossary expansion, semantic retrieval using SBERT, metadata enrichment via LLM-generated data product descriptions, synthetic query generation, model fine-tuning, and LLM-based reranking. Each step is evaluated using consistent, industry-aligned metrics, enabling direct comparison across techniques.

The experimental results demonstrate that the transition from lexical to semantic retrieval yields the largest observed performance gains. In particular, semantic retrieval improves overall discoverability by approximately 25–30%, with top-5 coverage increasing from roughly two thirds of queries to over 80%. Subsequent metadata enrichment techniques provide incremental improvements, particularly in precision, while more advanced approaches such as fine-tuning and reranking exhibit diminishing returns and strong dependence on data quality. However, some techniques lead to reduced performance, highlighting pitfalls and limitations.

By grounding these findings in a real-world enterprise dataset and focusing exclusively on metadata-level information, this study provides practical insights into how organizations can improve data discoverability without relying on extensive manual annotation or access to underlying data. The proposed framework is designed to be adaptable to different organizational contexts and data maturity levels, offering a pragmatic pathway toward more effective, AI-driven metadata management.

Although the experimental evaluation focuses on a single enterprise domain, the proposed framework is intentionally model-agnostic and designed to be applicable across different domains with minimal adaptation.

2 Literature Review

The enrichment of metadata for Data Products remains a relatively underexplored research area, primarily because the concept of Data Products itself is still emerging and has not yet reached wide industrial maturity [4]. As a result, direct studies focusing explicitly on metadata enrichment for Data Products are limited. Nevertheless, a substantial body of

related work exists in adjacent domains, particularly in semantic retrieval, metadata enrichment, and the application of Large Language Models (LLMs) to structured and semi-structured data.

Based on a focused review of the literature, we identify five categories of techniques that are most relevant to metadata-driven data discovery. These categories form the conceptual basis for the experimental framework proposed in this study. Figure 1 provides a conceptual grouping of AI-based retrieval and metadata enrichment approaches identified in the literature, which informs the design and scope of the experimental framework.

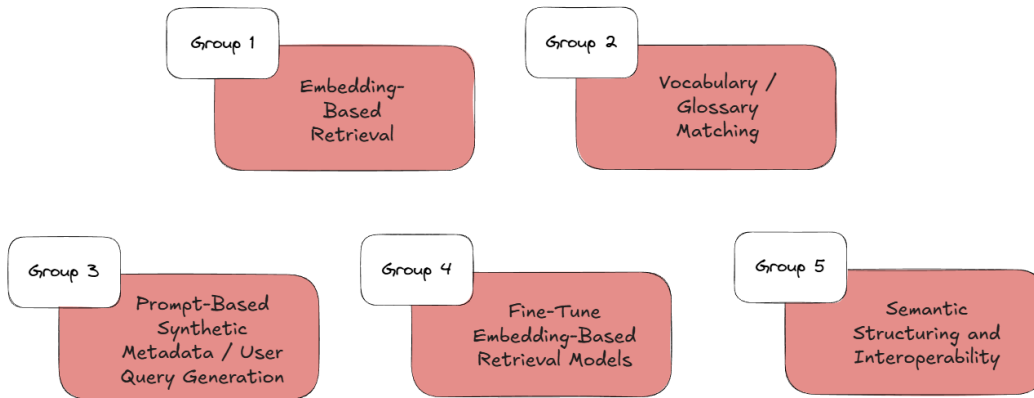


Figure 1 Conceptual grouping of AI-based retrieval and metadata enrichment approaches identified in the literature. The proposed experimental framework evaluates four of the five identified groups, while semantic structuring and interoperability are included for conceptual completeness but are not evaluated experimentally.

2.1 Embedding-Based Retrieval

In the data retrieval area, several techniques have been implemented. One particularly effective and simple technique for our task is the use of Sentence-BERT (SBERT). SBERT is an extension of the original BERT architecture, designed specifically for semantic similarity tasks (it can “understand the context”), which are crucial to identifying similar Metadata entries or matching user queries to Data Product descriptions. Unlike traditional BERT, which requires both sentences to be processed together in a cross-encoder setup (resulting in high computational cost), SBERT uses a Siamese network structure to generate fixed-size embeddings for individual sentences independently. These embeddings can then be compared efficiently using cosine similarity. As Nils Reimers et al. [5] showed, SBERT significantly outperforms BERT in both accuracy and speed, reducing the time needed for similarity search from hours to seconds. This makes SBERT a powerful tool for building scalable, reliable and cost-efficient retrieval systems. Naofumi Fujishiro et al. [6] also

demonstrated the strong performance of SBERT in retrieval tasks, particularly in contexts comparable to data product retrieval (for example, in their classification of accident reports).

2.2 Vocabulary and Glossary Matching (Expander)

Several studies focus on aligning technical metadata with business-oriented vocabularies to improve data discovery. Lobo et al. [7] explored matching column names with business glossary terms in scenarios where only minimal metadata were available and no training data existed. By leveraging LLMs to generate additional column context and identify relationships with glossary terms, their approach improved dataset retrieval without manual intervention.

Martorana et al. [8] adopted a similar strategy using a controlled vocabulary from the Consortium of European Social Science Data Archives (CESSDA), operating in a zero-shot setting by integrating domain-specific topics directly into LLM prompts. While effective, this approach depends on the availability of well-defined controlled vocabularies. Sundaram et al. [9] further demonstrated that combining LLMs with structured knowledge schemas, such as ontologies provided by the CEDAR Workbench, significantly improves metadata quality compared to using LLMs alone.

Singh et al. [10] proposed a metadata enrichment pipeline centered on an “Expander” component that enhances column and table descriptions using curated metadata and business glossaries. By combining pretrained and fine-tuned models trained on large curated datasets, their approach achieved high approval rates from data stewards, highlighting the practical effectiveness of glossary-driven metadata expansion.

2.3 Prompt-Based Metadata and Synthetic Query Generation

Several studies explore the use of LLMs to generate synthetic queries and enrich metadata for improved data discovery. Silva et al. [11] proposed the DAPDR pipeline, which generates synthetic queries from dataset titles, descriptions, and keywords, reducing the need for manual annotation such as business glossaries. Their results showed that synthetic queries are particularly useful when real user queries are sparse or unavailable. Similar findings were reported by Jeronymo et al. [12], who focused on generating diverse and semantically meaningful synthetic queries for retrieval tasks.

The role of prompt engineering in retrieval and metadata enrichment has been examined extensively. Zhu et al. [13] demonstrated that carefully designed prompts can significantly

improve synthetic query generation, document ranking, and metadata enrichment, especially in low-resource settings. Liu et al. [14] provided a systematic overview of prompt-based learning, showing that prompting can support a wide range of downstream tasks without requiring model fine-tuning, making it particularly effective in zero- and few-shot scenarios.

The importance of prompt engineering is further highlighted by Polak and Morgan [15], who showed that conversational language models combined with well-designed prompts can accurately transform unstructured text into structured data, reinforcing the value of prompt-based approaches for metadata generation and enrichment.

2.4 Fine-Tuning Embedding-Based Retrieval Models

Building on previous findings, Nils Reimers et al. [5] showed that fine-tuning SBERT can lead to significantly better results. Similarly, Nandan Thakur et al. [16], evaluated the performance of a fine-tuned SBERT model across different datasets, using the BEIR benchmark. BEIR (Benchmarking Information Retrieval) is a standardized framework for evaluating and comparing retrieval models, particularly neural and embedding-based ones such as SBERT, DPR, and ColBERT. In most cases, SBERT overperformed similar methods, especially when fine-tuned on domain-relevant datasets.

These results highlight the importance of task-specific fine-tuning in retrieval contexts. While SBERT performs well in general, adapting it to the characteristics of a specific domain or dataset can significantly enhance accuracy and relevance in retrieval tasks. Fine-tuning aligns the model more closely with the vocabulary, structure, user queries and semantics of the target data, which is especially beneficial when working with specialized or sparse Metadata.

2.5 Semantic Structuring and Interoperability

More advanced approaches to metadata enrichment focus on explicitly modeling semantic relationships. Peng et al. [17] explored the use of GraphRAG, combining retrieval with structured graph representations to capture relational knowledge. While effective for complex tasks such as question answering, this approach requires the availability of knowledge graphs and supporting infrastructure, which can be costly and difficult to maintain in enterprise environments.

Mombaerts et al. [18] proposed a framework centered on meta-knowledge structures for improving retrieval in RAG systems. By classifying metadata into categories such as summaries, schemas, and intent, their approach demonstrates how synthetic metadata can support more systematic organization and retrieval.

Standardized metadata schemas and semantic models have also been proposed as a foundation for interoperability. Baker et al. [19] highlighted the use of RDF-based application profiles aligned with established vocabularies such as Dublin Core [20] to define machine-understandable semantic relationships. While these approaches support interoperability and long-term consistency, they often rely heavily on manual curation.

Finally, ISO/IEC 11179-1:2023 [21] provides guidance on metadata management best practices, offering a reference model for defining data elements that support interoperability, reuse, and semantic clarity. Together, these approaches illustrate the trade-offs between expressive semantic modeling and the practical cost and effort required for implementation.

2.6 Positioning of This Study

Taken together existing research provides a diverse set of techniques for improving metadata-driven retrieval. However, most prior work evaluates these techniques in isolation or focuses on specific components of the discovery pipeline. In contrast, this study systematically evaluates embedding-based retrieval, metadata enrichment, synthetic data generation, fine-tuning, and reranking within a unified, step-by-step experimental framework, with a specific focus on practical applicability in enterprise data product discovery.

3 Methodology and Experimental Framework

3.1 Research Design Rationale

The primary objective of this study is to evaluate how different Artificial Intelligence techniques can improve data product discoverability through metadata enrichment. Rather than proposing a single method or model, the research adopts an incremental experimental design that reflects how organizations typically evolve their data discovery capabilities in practice.

This step-by-step approach is motivated by two observations from both industry practice and prior research. First, organizations rarely transition directly from simple keyword-based search to fully AI-driven solutions. Instead, they progressively introduce more advanced techniques as data maturity, tooling, and skills improve. Second, existing studies often evaluate individual techniques in isolation, making it difficult to understand their relative contribution and interaction effects when combined within a single pipeline.

To address these gaps, the experimental framework is designed as a sequence of increasingly sophisticated retrieval and enrichment steps. At each step, performance is measured using the same evaluation metrics, enabling direct comparison and attribution of performance changes to specific techniques. While semantic structuring and interoperability approaches are discussed in the literature and included in the conceptual grouping, they are not incorporated into the experimental evaluation due to scope and data availability constraints.

3.2 Dataset Description

The experiments are conducted on a real-world enterprise dataset consisting exclusively of metadata describing 30 data products. The dataset originates from an energy-sector use case and reflects typical metadata available in enterprise data catalogs. For confidentiality reasons, system names, identifiers, and business-specific terminology have been anonymized.

Each data product is described using a set of basic metadata attributes, including:

- Data product name
- Short textual description
- Column names (schema-level metadata)
- Associated tags

Importantly, no actual data records are used in the experiments. All retrieval tasks operate solely on metadata, reflecting a common enterprise constraint where access to underlying data is restricted or unnecessary for discovery purposes.

In addition to data product metadata, a set of 38 labeled user queries (real user queries) is used as ground truth. Each query is manually associated with the correct data product. These queries originate from a production data discovery solution and represent real information needs expressed by users. Their quality has been validated by domain experts, including data teams, data stewards, and data users. While limited in scale, this dataset

represents high-confidence, expert-validated ground truth from a production environment, prioritizing qualitative accuracy over synthetic scale.

While the dataset size is relatively small, it enables controlled experimentation and detailed analysis of how different techniques affect discoverability. As later sections show, even within this limited setting, substantial performance differences emerge between retrieval approaches. Although limited in size, the dataset reflects realistic enterprise constraints, where metadata is often sparse, access to underlying data is restricted, and labeled user queries are limited.

3.3 Evaluation Metrics

To assess retrieval performance, three complementary metrics are used consistently across all experiments: Total Score, Top-1 Accuracy, and Top-5 Coverage.

- **Top-1 Accuracy** measures precision by evaluating how often the correct data product appears as the top-ranked result.
- **Top-5 Coverage** measures recall-oriented performance by evaluating how often the correct data product appears within the top five recommendations.
- **Total Score** is used to capture overall ranking quality by assigning position-dependent weights to the rank at which the correct data product appears. Formally, for a set of queries Q , the Total Score is defined as:

$$Score = \sum_{q \in Q} w(r_q)$$

where r_q denotes the rank position of the correct data product for query q , and the weighting function $w(r)$ is defined as:

$$w(r) = \begin{cases} (k + 1) - r, & \text{if } 1 \leq r \leq k \\ 0, & \text{otherwise} \end{cases}$$

In this study, the evaluation window is set to $k = 5$, corresponding to the top-5 results returned by the retrieval system. The weighting function therefore assigns 5 points to a top-1 match, decreasing linearly to 1 point for a rank-5 match, and 0 points if the correct data product does not appear within the top-5 results.

To ensure comparability with existing literature, we align our evaluation with commonly used retrieval settings and rank-based metrics. In particular, we focus on Top-1 Accuracy and Top-5 Coverage, which capture precision- and recall-oriented performance, respectively.

Additionally, to reflect the practical utility for enterprise users, we introduce a task-specific “Total Score” metric that assigns linear weights to the top-5 results, balancing methodological rigor with practical interpretability. In enterprise data discovery settings, practitioners typically focus on whether the correct dataset appears within the first few results rather than on logarithmic discounting schemes. The chosen metric exhibits behavior comparable to a simple linearly discounted ranking function while remaining intuitive and actionable for real-world decision making.

Together, the three metrics provide a balanced evaluation of retrieval performance, capturing precision (Top-1 Accuracy), coverage (Top-5 Coverage), and ranking quality (Total Score).

3.4 Step-by-Step Experimental Framework

The experimental framework consists of seven main steps, each building on the output of the previous one. This design enables systematic evaluation of both individual techniques and their cumulative impact. The step-by-step experimental framework is illustrated in Figure 2.

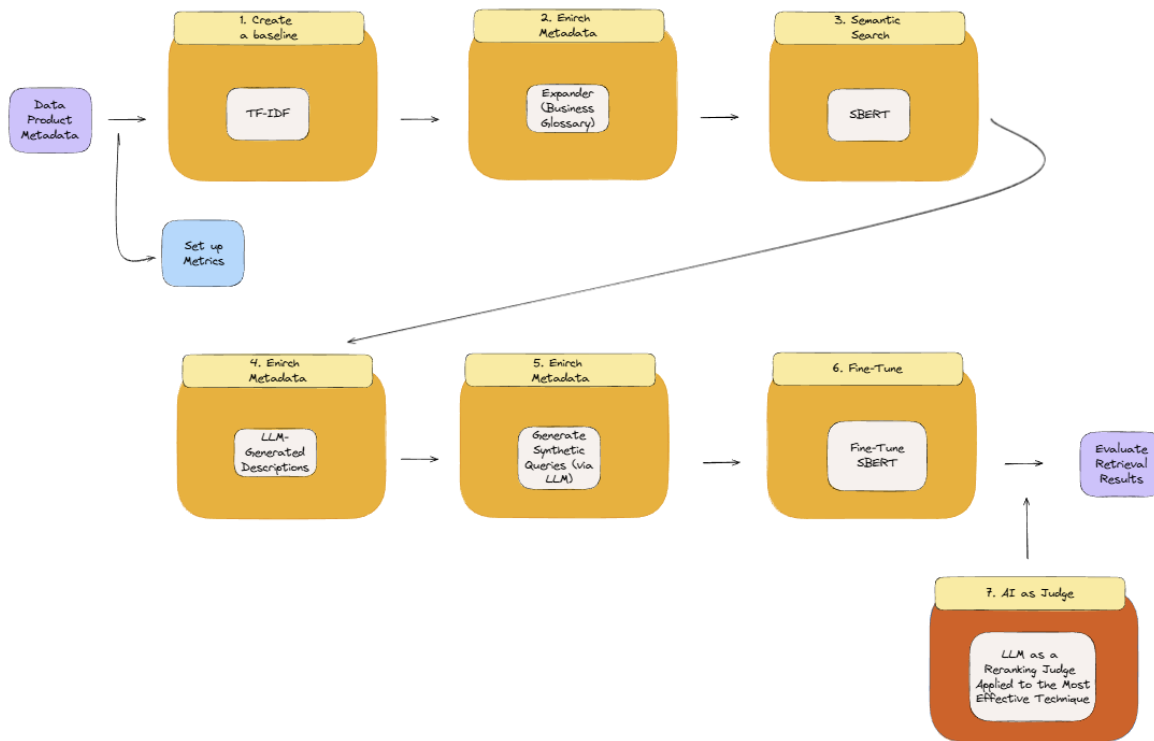


Figure 2 Step-by-step experimental framework illustrating the seven-stage retrieval and metadata enrichment pipeline evaluated in this study. Each step incrementally builds on the previous one, starting from a keyword-based baseline and progressively

introducing metadata enrichment, semantic retrieval, synthetic query generation, model fine-tuning, and LLM-based reranking, with performance evaluated consistently across all stages.

3.4.1 Step 1: Keyword-Based Baseline (TF-IDF)

The first step establishes a baseline using the TF-IDF keyword-based retrieval method [22]. All available metadata attributes for each data product are concatenated into a single textual representation. User queries are processed using the same vectorization approach, and cosine similarity is used to rank data products.

This baseline serves as a reference point for all subsequent experiments. As the results demonstrate, while TF-IDF performs reasonably well in some cases, it struggles to consistently surface relevant data products, particularly when user queries and metadata differ in terminology.

3.4.2 Step 2: Metadata Enrichment via Business Glossary Expansion

In the second step, metadata is enriched using a business glossary expansion mechanism. Technical or abbreviated column names are mapped to more descriptive, business-oriented terms. The enriched metadata is then used as input to the same TF-IDF retrieval process.

This step evaluates whether improving metadata quality alone, without changing the retrieval model, can enhance discoverability. As later results show, glossary expansion leads to modest but consistent improvements, particularly in precision.

3.4.3 Step 3: Semantic Retrieval with Sentence-BERT

The third step replaces keyword-based retrieval with semantic embedding-based retrieval using Sentence-BERT (SBERT). Data product metadata and user queries are encoded into dense vector representations, and similarity is computed using cosine similarity.

Multiple SBERT variants are evaluated to assess the impact of model architecture and training objectives. The results demonstrate that semantic retrieval represents a major performance shift, yielding the largest improvement across all metrics compared to the baseline.

3.4.4 Step 4: LLM-Generated Data Product Descriptions

In the fourth step, Large Language Models are used to generate enhanced data product descriptions based on existing metadata. These descriptions are designed to be more user-friendly and descriptive, aiming to improve semantic alignment between queries and metadata.

The generated descriptions are incorporated into the metadata representation and evaluated as part of the semantic retrieval process. This step tests whether LLM-generated metadata can compensate for sparse or technical descriptions.

3.4.5 Step 5: Enhance Metadata with Synthetic Query Generation

The fifth step introduces synthetic user queries generated by an LLM. These queries are intended to approximate how users might search for each data product and are used both as additional metadata enrichment signals and as training data for fine-tuning.

This step explores the feasibility of using synthetic data to improve retrieval performance in scenarios where real user queries are limited or unavailable.

3.4.6 Step 6: Fine-Tuning Semantic Models with Synthetic and Real User Queries

In the sixth step, SBERT models are fine-tuned using two alternative training sets: synthetic queries and real user queries. This enables a direct comparison of the impact of query quality on model adaptation and retrieval performance.

As the results later show, fine-tuning can yield improvements, but only when high-quality, representative queries are used.

3.4.7 Step 7: LLM-Based Reranking (AI as a Judge)

The final step applies an LLM-based reranking mechanism to the top results produced by the best-performing retrieval configuration. Rather than replacing the retrieval model, the LLM acts as a post-processing judge that evaluates and potentially reorders the top recommendations.

This step assesses whether reasoning-based refinement can further improve precision without substantially increasing system complexity.

3.5 Summary

The proposed experimental framework provides a structured and incremental approach to evaluating AI-driven metadata enrichment techniques. By measuring performance at each step using consistent metrics, the framework enables clear attribution of improvements and highlights trade-offs between effectiveness, complexity, and data quality.

The next section presents the experimental results for each step, providing a detailed analysis of how performance evolves as increasingly sophisticated techniques are introduced.

4 Experimental Results and Evaluation

This section presents the experimental results obtained by applying the proposed framework. The objective is to analyze how retrieval performance evolves as increasingly sophisticated metadata enrichment techniques are introduced and to quantify the contribution of each step using consistent evaluation metrics.

All results are reported using Total Score, Top-1 Accuracy, and Top-5 Coverage, as defined in Section 3.3. Rather than repeating all numerical values from tables, the analysis focuses on relative improvements, performance trends, and practical implications.

4.1 Baseline Performance: TF-IDF

The keyword-based TF-IDF baseline establishes the reference point for the entire study. Using only the available metadata without enrichment, the baseline achieves a Total Score corresponding to approximately **59%** of the maximum achievable score. In terms of precision, the correct data product appears as the top-ranked result in **50%** of user queries (in one out of every two queries). Furthermore, the correct data product is included within the top five recommendations in approximately **66%** of the cases.

While these results indicate that TF-IDF can retrieve relevant results in some scenarios, they also reveal substantial limitations. In roughly one third of the queries, the correct data product does not appear in the top five results at all. From a user perspective, this level of coverage is likely to lead to frustration and reduced trust in the discovery system. These findings confirm that lexical matching is insufficient for supporting reliable data discovery in enterprise environments characterized by heterogeneous metadata and varied user language.

4.2 Effect of Business Glossary Expansion (Expander)

The introduction of business glossary expansion leads to consistent, though moderate, improvements over the TF-IDF baseline. By enriching column names with descriptive business terms, vocabulary mismatches between metadata and user queries are reduced. For example, the word “eta” expanded to “estimated time of arrival”. In our study, we reviewed 2.835 column names and expanded 1.624 of them using the Business Glossary.

Empirically, glossary expansion leads to overall improvements, although the gains remain modest. The largest impact is observed in Top-1 Accuracy, with an increase of approximately **2.5%** (from 50,0% to 52,5%). This indicates that the effect of glossary expansion is limited when applied in combination with keyword-based retrieval. Although coverage improves slightly, a non-negligible portion of queries still fails to retrieve the correct data product

within the top five results. Overall, these findings suggest that metadata enrichment is beneficial but cannot fully compensate for the inherent limitations of lexical retrieval models, which are unable to capture the semantic meaning of the content.

4.3 Transition to Semantic Retrieval with SBERT (+ Expander)

Enhancing Expander with semantic retrieval using Sentence-BERT represents the largest performance shift across all experiments. When applied to the same metadata, SBERT-based retrieval (keeping Expander) improves the Total Score by approximately **16,0%** relative to the case that we use only Expander (4.2).

Top-1 Accuracy increases from **52,5%** to **63%**, while Top-5 Coverage rises sharply from approximately **66%** to over **84%**. In practical terms, this means that in more than four out of five user queries, the correct data product is surfaced within the top five recommendations.

These results confirm that semantic understanding is a critical factor in enterprise data discovery. SBERT's ability to capture contextual meaning allows it to match queries and metadata even when there is little or no lexical overlap. Notably, experiments with multiple SBERT variants reveal that **model purpose and training objectives** are more influential than model size. Smaller models optimized for question–answering and semantic similarity outperform larger, more generic embeddings in this task.

In our case, the best-performing model was multi-qa-MiniLM-L6-cos-v1, which contains approximately 22.7 million parameters. Despite its relatively small size, this model outperformed larger alternatives such as all-mpnet-base-v2, which contains approximately 109 million parameters. This behavior can be attributed to differences in training objectives and embedding geometry rather than model capacity. The multi-qa-MiniLM-L6-cos-v1 model is explicitly fine-tuned for question–answer semantic similarity, which closely matches the structure of enterprise search queries that often resemble implicit questions. In contrast, larger general-purpose embedding models are optimized for broader semantic representation, which may introduce noise in metadata-driven retrieval tasks characterized by short, technical, and sparse textual descriptions. As a result, task-aligned models with fewer parameters can produce more discriminative embeddings in constrained enterprise search settings.

4.4 Impact of LLM-Generated Data Product Descriptions (SBERT + Expender + Synthetic Queries)

In this step, Large Language Models (LLMs) were used to generate enriched, user-friendly data product descriptions through prompt engineering. The generated descriptions were integrated into the metadata and evaluated within the existing SBERT-based retrieval

framework, allowing direct comparison with the previous configuration using SBERT with glossary expansion only.

Specifically, we used the GPT-3.5-turbo model for cost-performance considerations, accessed via the OpenAI API. Through prompt engineering, the LLM was assigned the role of a data catalogue assistant and instructed to generate a concise, 3–4 sentence description of each dataset, explaining its contents and intended use in clear, business-friendly language.

In addition, the experiment was repeated ten times to account for variability in the generated descriptions, as the temperature was set to 0.3. The final results report the average value of each evaluation metric across these runs. The temperature was intentionally kept low to balance linguistic diversity with semantic stability, allowing the model to produce slightly varied yet consistent descriptions without introducing excessive randomness.

The results show that incorporating LLM-generated data product descriptions consistently improves retrieval performance across all evaluation metrics. Compared to the previous step, Total Score, Top-1 Accuracy, and Top-5 Coverage **all increased by 1%, 5% and 2,5% respectively**, indicating that the enriched descriptions provide additional semantic context that improves alignment between user queries and metadata.

These improvements suggest that natural language descriptions generated by LLMs enhance the expressiveness and clarity of metadata, enabling the SBERT model to better capture underlying semantic relationships. Unlike later enrichment steps, this approach delivers measurable performance gains without requiring human intervention or query-level validation, highlighting its scalability and practical value.

Overall, this experiment demonstrates that LLM-generated data product descriptions represent an effective and low-effort metadata enrichment strategy, contributing to the cumulative performance improvements observed across the framework.

4.5 Enrich Metadata with Synthetic Query Generation

Synthetic user queries were generated using an LLM (GPT-3.5-turbo, as in Step 4.4) and incorporated into the metadata of each data product as an additional enrichment signal. For each data product, three synthetic queries were automatically generated based on the LLM-generated descriptions, following the same fully automated process as in previous enrichment steps. The LLM was assigned the role of a search assistant helping users discover datasets in a data catalog. The experiment was repeated ten times, following the same methodology as in Step 4.4, to account for variability in the generated queries and to report averaged performance metrics.

The results show that enriching metadata with synthetic queries did not improve retrieval performance. Compared to the configuration using SBERT with glossary expansion and LLM-generated data product descriptions (4.3), the inclusion of synthetic queries led to a **decline** in Total Score **(-4%)** and Top-1 **(-5,5%)** Accuracy, while Top-5 Coverage remained unchanged.

The introduction of synthetic queries did not improve retrieval performance and, in fact, resulted in worse outcomes. This behavior can be attributed to several factors. The generated queries may not accurately reflect how end users formulate search requests, leading to a mismatch with real user intent, while the assignment of each synthetic query to a single data product introduces ambiguity in cases where product boundaries are not clearly defined. In addition, the substantial gains achieved in earlier steps suggest that the SBERT model may have reached a saturation point, making further automated enrichment less effective. The key insight from this experiment is that more metadata is not necessarily better: although synthetic queries are scalable, their effectiveness depends on strong domain alignment and semantic quality. At this stage of the framework, further performance improvements are more likely to require human-in-the-loop validation or complementary reasoning-based techniques, rather than additional automated metadata generation alone.

4.6 Fine-Tuning SBERT Models with Synthetic Queries & User Queries

In this step, the model is fine-tuned implementing two different approaches: fine-tuning using LLM-generated synthetic queries and fine-tuning using real user queries. The objective is to assess how query quality influences the effectiveness of task-specific model adaptation.

In the first case, SBERT (multi-qa-MiniLM-L6-cos-v1) was fine-tuned using the synthetic queries generated in Section 4.5. Different training configurations were evaluated by varying the number of epochs (1 and 5), and the experiment was repeated ten times, consistent with previous steps. The best performance was achieved with a single training epoch. A likely explanation is that increasing the number of epochs leads to overfitting synthetic queries, which are generated by an LLM and may not accurately represent real user behavior. As the model increasingly adapts its weights to these synthetic patterns, its ability to generalize to real user queries may deteriorate.

Under this process, both Total Score and Top-1 Accuracy **declined by 4% and almost 10% respectively** compared to the results in 4.4, while Top-5 Coverage remained unchanged. These results suggest that fine-tuning on synthetic queries may negatively affect retrieval performance when the generated queries do not accurately reflect real user intent, leading to suboptimal adjustments of the embedding space.

In the second case, the model was fine-tuned using real user queries. For this purpose, 24 queries were used as the training set and 14 queries as the test set. As in previous steps, the experiment was repeated ten times to smooth out the non-deterministic nature of the process.

In contrast to synthetic queries, fine-tuning with real user queries produced the best overall results across the entire experimental framework. All evaluation **metrics improved compared to 4.4 by almost 1%**, indicating that high-quality, domain-grounded queries enable more effective model adaptation. By learning directly from authentic user behavior, the fine-tuned model better captures domain-specific terminology, intent, and search patterns.

Overall, these findings demonstrate that fine-tuning effectiveness is highly dependent on query quality. While fine-tuning can substantially enhance semantic retrieval, it is beneficial only when the training signals are representative of real user intent. Fine-tuning on low-quality or weakly grounded synthetic queries may instead degrade performance, reinforcing the need for careful selection and validation of training data in domain-adaptive retrieval systems.

4.7 LLM-Based Reranking (AI as a Judge)

As a final enhancement, an LLM-based reranking mechanism is applied to the best-performing retrieval configuration, namely the fine-tuned SBERT model (Section 4.6, trained with real user queries). In this step, the LLM (GPT-3.5 Turbo) acts as a “judge” by evaluating and potentially reordering the top five data product recommendations returned by SBERT for a given user query, with the objective of correcting ranking errors without modifying the underlying retrieval model. In practice, the five recommendations produced by SBERT are passed to the LLM, which is instructed to adjust the ranking only when necessary based on its judgment.

Initial experiments showed that a direct zero-shot prompting application of the LLM-based judge did not improve retrieval performance and, in some cases, resulted in performance degradation. Subsequent error analysis revealed that both SBERT and the LLM struggled with longer and more complex queries involving multiple constraints or comparative conditions. Specifically, an analysis of successful versus failed cases showed a notable difference in query length, with failed queries averaging 146 characters compared to 97 characters for successful ones.

Based on these findings, the reranking strategy was refined through an iterative prompt refinement process, incorporating explicit guidance tailored to query complexity. Shorter queries were handled using simpler instructions, while longer and more complex queries

were addressed with more structured prompts designed to better guide the LLM’s reasoning behavior.

With this revised setup, and after multiple trial-and-error iterations, the AI-as-judge approach successfully outperformed the fine-tuned SBERT model described in Section 4.6 (user queries). After adjustment to a common evaluation scale, Total Score **increased from approximately 79% to 84%, while Top-1 Accuracy improved from 69% to 78%**. Top-5 Coverage remained unchanged compared to the previous step. These results indicate that reasoning-based reranking can meaningfully improve precision by correcting top-ranked errors that embedding-based similarity alone fails to resolve.

Overall, this experiment demonstrates that LLM-based reranking can deliver substantial gains when carefully designed, but it is not a plug-and-play solution. Its effectiveness depends on prompt design, error analysis, and explicit modeling of query complexity, making it a powerful yet demanding technique in terms of both time and expertise for incremental performance improvement.

5 Discussion

5.1 Overview of Findings

This section discusses the experimental results presented in Section 4 and interprets their implications for enterprise data product discovery. Overall, the results indicate that the transition from lexical to semantic retrieval represents the most impactful change across the entire experimental framework. Metadata enrichment techniques contribute incremental improvements, while more advanced approaches such as fine-tuning and LLM-based reranking introduce diminishing returns and increased sensitivity to data quality.

These trends are visually summarized in Figure 3, which illustrates the non-monotonic performance evolution observed as increasingly advanced metadata enrichment and model adaptation techniques are introduced.

Notably, several techniques led to reduced performance under certain conditions, highlighting that increasing system sophistication does not necessarily translate into better retrieval outcomes. These findings emphasize the importance of selecting techniques that are aligned with both the characteristics of the metadata and the maturity of the organizational context.

Performance Evolution Across Metadata Enrichment Techniques

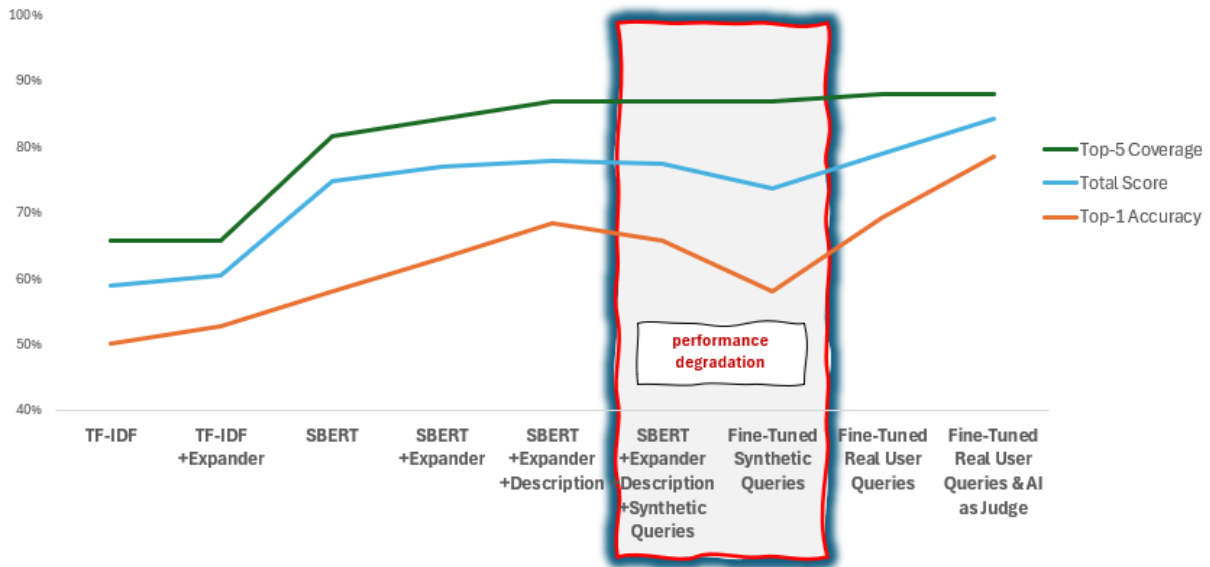


Figure 3 Performance evolution across successive metadata enrichment and retrieval configurations. The figure shows changes in Total Score, Top-1 Accuracy, and Top-5 Coverage as increasingly advanced techniques are introduced. While semantic retrieval and metadata enrichment lead to substantial improvements, certain configurations involving synthetic queries and fine-tuning exhibit stagnation or performance degradation, highlighting sensitivity to data quality and training signal characteristics.

5.2 Why Semantic Retrieval Dominates

The strong performance gains observed when transitioning from keyword-based to semantic retrieval can be attributed to the fundamental limitations of lexical matching in enterprise environments. Metadata descriptions are often short, technical, and heterogeneous, while user queries frequently vary in phrasing and intent. Under these conditions, lexical overlap between queries and metadata is inconsistent, limiting the effectiveness of keyword-based approaches.

Semantic embedding models such as SBERT address this limitation by capturing contextual meaning rather than surface-level token similarity. This allows semantically related concepts to be matched even in the absence of shared vocabulary. The results suggest that semantic alignment, rather than metadata volume alone, is the primary driver of improved discoverability in enterprise data catalogs.

5.3 Diminishing Returns of Advanced Techniques

While metadata enrichment and advanced modeling techniques provide additional performance gains, their impact is markedly smaller compared to the transition to semantic retrieval. LLM-generated data product descriptions improve semantic expressiveness and yield consistent but modest gains, particularly in precision. In contrast, synthetic query generation and fine-tuning demonstrate mixed results, with performance highly dependent on the quality and representativeness of the generated queries.

These findings illustrate a pattern of diminishing returns, where each additional layer of sophistication introduces greater complexity and sensitivity to noise. In some cases, automated enrichment techniques introduce semantic ambiguity that negatively affects retrieval performance. This suggests that beyond a certain point, further automation requires either human validation or complementary reasoning-based mechanisms to maintain reliability.

5.4 Impact of Dataset Size and Enterprise Constraints

The experimental evaluation is conducted on a relatively small dataset consisting of 30 data products and 38 expert-validated user queries. While this limits statistical generalization, the dataset reflects realistic enterprise constraints, where labeled queries and high-quality metadata are scarce. The consistent trends observed across multiple experimental steps indicate that the relative performance differences between techniques are robust within this context.

Importantly, the results should be interpreted as comparative rather than absolute performance claims. The proposed framework is designed to highlight relative gains, trade-offs, and failure modes when introducing AI-driven metadata enrichment techniques. Applying the framework to larger or cross-domain datasets represents a natural direction for future work.

5.5 Practical Implications for Enterprise Adoption

From a practical perspective, the findings support a pragmatic, incremental approach to AI adoption in metadata management. Organizations can achieve substantial improvements in data discoverability by prioritizing semantic retrieval and lightweight metadata enrichment before investing in more complex techniques such as fine-tuning or LLM-based reranking.

The results also demonstrate that model size alone is not a reliable indicator of performance. Instead, alignment between training objectives and the target retrieval task plays a more critical role. For many organizations, smaller, task-aligned models combined with high-

quality metadata may offer a more cost-effective and maintainable solution than larger, general-purpose alternatives.

An additional practical insight concerns model selection for semantic retrieval. The experimental results indicate that retrieval performance is not primarily driven by model size, but by alignment between the model's training objectives and the target retrieval task. In this study, a smaller SBERT variant fine-tuned for question-answer semantic similarity outperformed larger, more general-purpose embedding models. This suggests that fit-for-purpose models are often better suited for enterprise metadata retrieval, where queries resemble implicit questions and metadata descriptions are short and domain-specific. From an operational perspective, this finding supports the selection of task-aligned, lightweight models that offer improved efficiency, lower computational cost, and easier deployment without sacrificing retrieval quality.

6 Conclusions and Future Work

6.1 Conclusions

This study examined how artificial intelligence techniques can be applied incrementally to improve data product discoverability through metadata enrichment in enterprise environments. By proposing and evaluating a step-by-step experimental framework, the research provides empirical evidence on the relative impact, trade-offs, and limitations of different retrieval and enrichment techniques when applied under realistic organizational constraints.

The results show that the transition from lexical to semantic retrieval constitutes the most impactful improvement across the entire framework. Semantic embedding-based retrieval consistently outperforms traditional keyword-based methods, substantially increasing both precision and coverage in metadata-driven search. Lightweight metadata enrichment techniques, such as business glossary expansion and LLM-generated data product descriptions, further improve performance by enhancing semantic clarity and interpretability.

More advanced approaches, including synthetic query generation, fine-tuning, and LLM-based reranking, deliver mixed outcomes. While these techniques can provide additional gains under appropriate conditions, they also introduce diminishing returns and increased sensitivity to data quality, with some configurations leading to performance degradation. These findings highlight that increased system complexity does not automatically translate into improved retrieval effectiveness.

A key takeaway of this study is that effective enterprise data discovery depends more on task alignment than on model size. Smaller, fit-for-purpose models aligned with the characteristics of metadata-driven retrieval tasks can outperform larger, general-purpose alternatives, offering a more efficient and operationally viable solution.

Overall, the findings support a pragmatic approach to AI adoption in metadata management. Organizations can achieve meaningful improvements in data discoverability by prioritizing semantic retrieval and targeted metadata enrichment, while carefully evaluating the costs, risks, and expected benefits of more advanced techniques. The proposed framework offers a structured foundation for guiding such decisions across different levels of data maturity.

6.2 Future Work

Several directions for future research emerge from this study. First, applying the proposed framework to larger and more diverse datasets would enable more robust conclusions and support cross-domain validation. Second, incorporating human-in-the-loop validation for metadata enrichment and synthetic query generation could improve data quality and system reliability.

Further work could also explore the integration of structured knowledge representations, such as knowledge graphs, to capture relationships between data products and metadata elements. In addition, developing explainability mechanisms for semantic retrieval and LLM-based components could increase transparency and trust among business users.

By building on the framework and findings presented in this study, future research can further advance AI-driven metadata management and support more effective data discovery in enterprise environments.

7 Declarations

Data Availability

The data supporting the findings of this study are not publicly available due to confidentiality restrictions related to enterprise data governance. The study relies exclusively on metadata-level information with anonymized identifiers.

Competing Interests

The author declares that there are no competing interests.

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Ethics Approval

Not applicable.

Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

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